BLA-06 · MANAGEMENT, GOVERNANCE AND MIGRATION

Keeping Things Running, and Moving Data In

AWS Auto Scaling · Cost Explorer and Cost Anomaly Detection · Amazon Managed Grafana · AWS Systems Manager · DataSync and Transfer Family

PART 01

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AWS Auto Scaling

Add servers when you're busy, remove them when you're not.

Setting	Meaning
Minimum	The fewest servers you'll ever run
Desired	How many you want right now
Maximum	The most you'll ever run

Policy	Example
Target tracking	"Keep average CPU at 50%"
Scheduled	"Add servers at 9 a.m."
Predictive	Learns the daily pattern and scales ahead of it

💡 **AWS Auto Scaling** also scales ECS services, DynamoDB tables and Aurora replicas.

Watching the bill

	Cost Explorer	Cost Anomaly Detection
Looks	Backwards — and forecasts	For surprises
Shows	Spend by service, account or tag	Unusual jumps in spend
How	Charts and reports	Machine learning + email or SNS alerts

🏢 **A test script loops overnight calling Bedrock.** Anomaly Detection flags it the next morning — not at month-end.

Amazon Managed Grafana

Open-source Grafana dashboards — AWS runs the servers.

- 🔌 Pulls data from CloudWatch, Prometheus, X-Ray, OpenSearch and more
- 🔐 People sign in with IAM Identity Center

	Managed Grafana	QuickSight
For	Engineers watching systems	Business users looking at data
Shows	Live health, metrics, errors	Business numbers and trends

AWS Systems Manager

A toolbox for running servers — on AWS and on-premises.

Tool	What it does
Fleet Manager	See and manage all your servers in one place
Patch Manager	Keep servers patched on a schedule
Run Command	Run one command on many servers at once
Session Manager	Log in to a server without SSH
Parameter Store	Store settings and secrets

⚠️ **Every server needs the SSM Agent** and an IAM role that lets it talk to Systems Manager — or it won't show up.

Session Manager

A shell on your server — without SSH.

SSH	Session Manager
Open port 22	No inbound ports
Hand out SSH keys	No keys — IAM decides who
Bastion host for private servers	Works on private servers
Hard to audit	Every session logged to S3 or CloudWatch

> **Needs:** SSM Agent + instance role with **AmazonSSMManagedInstanceCore** + outbound HTTPS (internet or VPC endpoints)

Parameter Store

Settings and secrets kept outside your code.

Type	Example
String	us-east-1
StringList	en,fr,de
SecureString (KMS-encrypted)	A database password

📁 `/genai/prod/model-id` → `amazon.nova-pro-v1:0`
📁 `/genai/prod/temperature` → `0.2`
📁 `/genai/dev/model-id` → `amazon.nova-lite-v1:0`

💡 **Need automatic rotation?** Use Secrets Manager. For ordinary settings, Parameter Store's standard tier is free.

Moving data into AWS

	AWS DataSync	AWS Transfer Family
Moves	Files from your storage (NFS, SMB, HDFS, other clouds)	Files sent by people and partners
Into	S3, EFS, FSx	S3 or EFS
How	Agent near your data; scheduled; copies only changes	SFTP, FTPS, FTP or AS2 — partners keep their tools

🏢 **Every night,** DataSync copies new policy documents to S3 → the knowledge base syncs → the chatbot knows today's policies.

Key takeaways

1. Auto Scaling — set a minimum, a maximum and a target
2. Cost Explorer looks back; Anomaly Detection catches surprises
3. Grafana for engineers, QuickSight for the business
4. Session Manager — no open ports, no keys, every session logged
5. DataSync for your files, Transfer Family for partners' files

Next · Part 2: getting requests in, and keeping data close